

COLD STORAGE TEMPERATURE MONITORING

Deployment, Hosting Cost & Client Pricing Guide

Industrial telemetry dashboard · FastAPI + React + SQL · Docker

Prepared for: Project owner / reseller

Prices are approximate, mid-2026, and depend on region, scale and client.

1. Executive Summary

This document explains how to deploy the Cold Storage Monitoring system, the best and cheapest places to host it, the real monthly running cost, and a realistic price to charge a client for the software. The application is a production-grade, secured web platform — not a prototype — so it can be hosted on a single small server and still serve a full facility.

Bottom line:

- **Cheapest reliable hosting:** one small VPS running Docker Compose — roughly **\$5–\$12 / month** (about █450–█1,000) including domain.
- **Biggest cost saver:** switch the production database from SQL Server to PostgreSQL — same code, ~half the RAM, no licensing weight.
- **Suggested one-time price to client:** **\$3,500–\$8,000** (█3.0L–█6.5L) for the software + setup, plus an annual support contract.

2. What You Are Deploying (Stack)

The project is a single repository with three runtime parts plus a reverse proxy. Everything is already containerised, so a server only needs Docker installed.

Component	Technology	Role	Resource need
Frontend (web)	React + Vite, served by Nginx	Dashboard & admin console UI	Tiny (static files)
Backend (api)	Python FastAPI (Uvicorn)	Auth, data API, SSE, reports, ingestion	~0.5 vCPU, 256–512 MB
Database (db)	SQL Server (or PostgreSQL)	Rooms, sensors, readings, users, audit	Biggest: 2–4 GB RAM for SQL Server
Gateway (nginx)	Nginx + TLS	HTTPS, routes / to web and /api to api	Tiny

Because the frontend and API are served from one origin through Nginx, login cookies stay first-party and no extra CORS/cookie setup is needed in production.

3. Where to Deploy — Three Options

Pick by how much you want to manage yourself versus pay for convenience.

Option	What it is	Monthly cost	Effort	Best for
A. Single VPS (recommended)	One cloud server runs all containers via Docker Compose (Hetzner, Contabo, Lightsail, DigitalOcean).	\$5–\$12	Medium — you manage the server	One client, best price, full control
B. Managed PaaS	Frontend on Vercel/Netlify (free), API on Render/Railway/Fly, managed Postgres.	\$15–\$35	Low — platform manages servers	Fast setup, auto-scaling, less ops
C. Big Cloud (AWS/Azure/GCP)	VM/containers + managed SQL + load balancer. Azure suits SQL Server.	\$50–\$150+	High — needs cloud skills	Enterprise, multi-site, compliance

Recommendation: For a single facility/client, use **Option A — a Hetzner or Contabo VPS with Docker Compose**. It is the cheapest, gives you the whole stack on one box, and the project already ships the exact compose + Nginx files for it. Put Cloudflare (free) in front for DNS, TLS and basic DDoS protection.

4. Monthly Hosting Cost — Detail

Approximate provider pricing, mid-2026. ■ values use ~■84/\$.

Option A — Single VPS (cheapest)

Item	Example / spec	USD / mo	■ / mo
VPS — with PostgreSQL DB	Hetzner CX22 / Contabo — 2 vCPU, 4 GB RAM	\$5–\$7	■450–■600
VPS — if keeping SQL Server	4 vCPU, 8 GB RAM (SQL Server is RAM-hungry)	\$10–\$16	■850–■1,350
Domain name	yourdomain.com (billed yearly ~\$12)	~\$1	~■85
TLS / HTTPS	Let's Encrypt or Cloudflare	Free	Free
Cloudflare CDN/DNS	Free plan	Free	Free
Backups (optional)	Provider snapshots / object storage	\$1–\$3	■85–■250
TOTAL (with Postgres)		\$6–\$11	■500–■950

Option B — Managed PaaS

Item	Provider	USD / mo	■ / mo
Frontend hosting	Vercel / Netlify / Cloudflare Pages (free tier)	Free	Free
Backend API	Render Starter / Railway / Fly.io	\$7	~■600
Managed Postgres	Render / Neon / Supabase	\$0–\$19	■0–■1,600
Domain	Registrar	~\$1	~■85

Item	Provider	USD / mo	■ / mo
TOTAL		\$15-\$35	■1,300-■3,000

Note: a managed **SQL Server** (Azure SQL) starts higher than Postgres; this is the main reason to move the DB to PostgreSQL before going managed.

5. How to Deploy — Step by Step (Option A)

This is the exact path for the recommended single-VPS setup. The repository already contains `infra/docker-compose.prod.yml` and the Nginx config.

Step	Action
1. Get a server	Create a VPS (Ubuntu 22.04+, 2–4 GB RAM). Note its public IP.
2. Point a domain	Add an A record for your domain → server IP. (Cloudflare free works well.)
3. Install Docker	<code>curl -fsSL https://get.docker.com sh</code> — installs Docker + Compose.
4. Copy the project	Clone the repo or upload it to the server (e.g. <code>git clone</code> or <code>scp</code>).
5. Configure <code>.env</code>	<code>cp .env.example .env</code> then set strong values: API_SECRET_KEY (random 32+ chars), API_COOKIE_SECURE=true , API_ALLOWED_ORIGINS =your domain, DB password, and the bootstrap admin email/password.
6. Add TLS certs	Put certificate files in <code>infra/certs/</code> (Let's Encrypt via certbot, or use Cloudflare origin cert).
7. Build & start	<code>docker compose -f infra/docker-compose.prod.yml --env-file .env up -d --build</code>
8. Migrate & seed	<code>... run --rm migrate</code> then <code>... run --rm seed</code> (creates tables + first admin user).
9. Verify	Open <code>https://yourdomain</code> — login screen appears. Sign in with the bootstrap admin, then create users in Admin → Dashboard Access.
10. Harden	Change the admin password, enable a firewall (allow 80/443/22 only), schedule DB backups.

Cost-saving switch to PostgreSQL (optional but recommended)

The code uses SQLAlchemy + Alembic, so moving to Postgres is mostly a connection-string change. In `.env` set `DATABASE_URL=postgresql+asyncpg://user:pass@db:5432/cold_room` and replace the SQL Server service in the compose file with a `postgres:16-alpine` container. This drops the server RAM requirement from 8 GB to 2 GB and removes SQL Server's heavy image — cutting hosting cost by roughly half.

6. What to Charge the Client

This is custom industrial software with real security (hashed passwords, CSRF, rate-limit + lockout, RBAC), a live real-time dashboard, a full admin console, Excel reporting, data ingestion, and container deployment. It represents roughly **6–10 weeks of senior full-stack work**. Price on the **value** it delivers, not hours.

6.1 Price ranges by market

Market / client type	One-time build + license	Notes
India SMB / small client	■1.0L – ■3.5L (\$1,200–\$4,200)	Price-sensitive; keep scope tight
Gulf / enterprise (e.g. UAE industrial)	\$4,000 – \$12,000 (■3.4L–■10L)	Budgets are higher; emphasise reliability & security
US / EU freelance or agency	\$8,000 – \$25,000+	Full agency pricing

6.2 Recommended package (sell it in tiers)

Item	What the client gets	Suggested price
Software license + source	Full application, branded, source code handover, documentation.	\$3,500 – \$6,500
Deployment & setup	Install on the client's server/domain, TLS, first admin, data wiring.	\$500 – \$1,200
Annual support (AMC)	Bug fixes, updates, monitoring, ~18–20% of build value / year.	\$800 – \$1,500 / yr
Hosting (managed)	You run & manage the server; bill cost + management.	\$30 – \$60 / mo
Extra rooms / features	Per change-request or per additional site/facility.	Quote per scope

6.3 A clean number to quote

For a serious industrial client (the most likely buyer of this system), a strong, defensible quote is:

Line item	Price
One-time software + deployment	\$5,000 (approx. ■4.2 Lakh)
Annual support & updates (optional)	\$1,200 / year (approx. ■1.0 Lakh)
Hosting & management (optional)	\$45 / month (approx. ■3,800)

How to present it: lead with the \$5,000 one-time figure as the headline, then position support and hosting as “peace of mind” add-ons. If the client pushes back, reduce scope (fewer reports, no source handover) rather than discounting blindly. If it is a Gulf/enterprise buyer, start nearer **\$7,000–\$8,000** — the security and real-time features justify it.

7. Notes & Assumptions

- All hosting prices are approximate provider list prices for mid-2026 and vary by region and promotions.
- ■ conversions use ~■84 per US\$ — adjust to the current rate.

- Project pricing is guidance, not a fixed valuation; the real number depends on the client, region, exclusivity (resell rights), and support commitment.
- Keep one production server, one staging copy, and automated daily database backups regardless of which option you choose.
- Before going live: rotate all secrets, set `API_COOKIE_SECURE=true`, enable HTTPS, and restrict the server firewall.

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